

Equalisation

Joe Leacy

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1 Introduction

2 Adaptive Equalisation

Tracking fast-varying channel effects like PMD is done with a 2x2 MIMO equaliser that employs 4 finite impulse response (FIR) filters to give outputs

$$x_{\text{out}} = \mathbf{h}_{xx}^H \mathbf{x}_{\text{in}} + \mathbf{h}_{xy}^H \mathbf{y}_{\text{in}}, \quad y_{\text{out}} = \mathbf{h}_{yx}^H \mathbf{x}_{\text{in}} + \mathbf{h}_{yy}^H \mathbf{y}_{\text{in}}. \quad (1)$$

The filter weights are updated using an error estimate as

$$\begin{aligned} \mathbf{h}_{xx} &\leftarrow \mathbf{h}_{xx} + \mu e_x \mathbf{x}_{\text{in}} x_{\text{out}}^*, & \mathbf{h}_{xy} &\leftarrow \mathbf{h}_{xy} + \mu e_x \mathbf{y}_{\text{in}} x_{\text{out}}^*, \\ \mathbf{h}_{yx} &\leftarrow \mathbf{h}_{yx} + \mu e_y \mathbf{x}_{\text{in}} y_{\text{out}}^*, & \mathbf{h}_{yy} &\leftarrow \mathbf{h}_{yy} + \mu e_y \mathbf{y}_{\text{in}} y_{\text{out}}^*. \end{aligned} \quad (2)$$